THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 7: **Probability** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A letter is chosen at random from the word EXAMINATION. The probability that it is a vowel is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{6}{11}$ **(c)** $\frac{7}{11}$ **(d)** $\frac{5}{11}$

2. In a town of 6000 people the probability that a randomly chosen person owns a smartphone is $\frac{2}{3}$. The expected number of owners is: 1

(a) 3999 **(b)** 4002 **(c)** 4001 **(d)** 4000

3. If $P(E) = 11/20$, then the probability that E does not happen is: 1

(a) $\frac{3}{7}$ **(b)** $\frac{9}{20}$ **(c)** $\frac{11}{20}$ **(d)** $\frac{1}{2}$

4. Three coins are tossed. The probability of getting at least one tail is: 1

(a) $\frac{3}{4}$ **(b)** $\frac{1}{8}$ **(c)** $\frac{7}{8}$ **(d)** $\frac{7}{9}$

5. Two dice are rolled together. The probability that the sum is 11 is: 1

(a) $\frac{1}{18}$ **(b)** $\frac{1}{19}$ **(c)** $\frac{17}{18}$ **(d)** $\frac{1}{9}$

6. Two dice are rolled together. The probability of getting the same parity on both dice is: 1

(a) $\frac{2}{3}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{1}{4}$

7. A die is rolled. The probability of getting a perfect square or a number at least 5 is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{3}{4}$

8. A bag has 2 red and 8 blue balls. After adding 5 more red balls, the probability of drawing a red ball is: 1

(a) $\frac{7}{15}$ **(b)** $\frac{7}{16}$ **(c)** $\frac{2}{5}$ **(d)** $\frac{8}{15}$

9. A letter is chosen at random from the word PROBABILITY. The probability that it is the letter B or I is: 1

(a) $\frac{4}{11}$ **(b)** $\frac{7}{11}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{5}{11}$

10. In a town of 15000 people the probability that a randomly chosen person owns a smartphone is $\frac{4}{5}$. The expected number of owners is: 1

(a) 12002 **(b)** 12001 **(c)** 12000 **(d)** 11999

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The probability of winning a game is $\frac{5}{12}$. Find the probability of not winning, and the expected number of wins in 240 games. 2

12. Two dice are rolled together. Find the probability that the sum of the numbers is 3. 2

13. A bag has 9 red and 6 blue balls. 6 more red balls are added. Find the new probability of drawing a red ball. 2

14. The probability of winning a game is $\frac{4}{9}$. Find the probability of not winning, and the expected number of wins in 360 games. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A die is rolled. Find the probability of getting 3
(i) a factor of 6,
(ii) a composite number,
(iii) a factor of 6 or a composite number.

16. In a survey of 400 people, 150 like tea only, 100 like coffee only and 60 like both. Find the probability that a person chosen at random likes tea (tea only or both), and the probability that the person likes neither (the rest). 3

17. A bag has 6 red and 12 blue balls. How many red balls must be added so that the probability of drawing a red ball becomes $\frac{3}{4}$? 3

OR

A die is rolled. Find the probability of getting 3
(i) a number not divisible by 3,
(ii) the number 5,
(iii) a number not divisible by 3 or the number 5.

18. In a survey of 160 people, 55 like tea only, 40 like coffee only and 25 like both. Find the probability that a person chosen at random likes tea (tea only or both), and the probability that the person likes neither (the rest). 3

OR

A bag has 4 red and 16 blue balls. How many red balls must be added so that the probability of drawing a red ball becomes $\frac{3}{5}$? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two dice are rolled together. Find the probability of 5
(i) a sum of 3,
(ii) a sum of 11,
(iii) a sum of 3 or 11.

OR

A factory tested 150 bulbs and found 6 defective. Find the probability that a bulb is defective, and estimate the number of defective bulbs in a lot of 3000. 5

20. Two dice are rolled together. Find the probability of 5
(i) a sum of 7,
(ii) a sum of 12,
(iii) a sum of 7 or 12.

OR

A factory tested 300 bulbs and found 9 defective. Find the probability that a bulb is defective, and estimate the number of defective bulbs in a lot of 10000. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND